「Section 4. 3

Let (X,d) be a Metric space, and let $f: X \to \mathbb{R}$ be a Continuous map.
Prove the zero set of f

$$Z(f) := \{x \in X \mid f(x) = 0\}$$

is closed.

proof. Let $y \in \overline{Z(f)}$ be given. That is, for any $r > 0$,

$$B_r(y) \cap Z(f) \neq \emptyset.$$

Given $\varepsilon > 0$, there exists $\delta > 0$ s.t

$$d(t,y) < \delta \Rightarrow |f(t) - f(y)| < \varepsilon$$

For fixed $\delta > 0$, Choose

$$x \in B_\delta(y) \cap Z(f)$$

Then $d(x,y) < \delta$ and $f(x) = 0$. Therefore

$$|f(x) - f(y)| = |f(y)| < \varepsilon$$

Since $\varepsilon > 0$ was arbitrary, $f(y) = 0$. That is, $y \in Z(f)$.

Alternatively, $Z(f) = f^{-1}[\{0\}]$.

「Section 4.2

Let X, Y be Metric spaces. If $f: X \to Y$ is Continuous, then for any $E \subset X$,

$$f[\overline{E}] \subseteq \overline{f[E]}.$$

proof. Let $f(x) \in f[\overline{E}]$ be given. Since f is Continuous, given $\varepsilon > 0$, there exists $\delta > 0$ s.t

$$t \in B_\delta(x) \Rightarrow f(t) \in B_\varepsilon(f(x))$$

Meanwhile, since $x \in \overline{E}$, $B_\delta(x) \cap E \neq \emptyset$

Chose $u \in B_\delta(x) \cap E$. Then, $f(u) \in B_\varepsilon(f(x))$ and $f(u) \in f[E]$. That is

$$f(u) \in B_\varepsilon(f(x)) \cap f[E] \neq \emptyset.$$

This implies $f(x) \in \overline{f[E]}$.

⌜Section 4.4

Let (X, d_X) and (Y, d_Y) be a Metric space, and $f, g : X \to Y$ be a Continuous map.
If $E \subseteq X$ is a dense in X, then $f[E]$ is dense in $f[X]$.
Moreover, If $f|_E = g|_E$, then $f = g$.

Proof. Since f is Continuous, $f[X] = f[\overline{E}] \subseteq \overline{f[E]}$ is clear.
Let $x \in X = \overline{E}$ be given. If $f(x) = g(x)$, there is nothing to prove.
Suppose $f(x) \neq g(x)$. Take $\varepsilon = d(f(x), g(x)) > 0$. Since f and g are Continuous, there exist $\delta > 0$ s.t

$$t \in B_\delta(x) \Rightarrow d(f(x), f(t)),\ d(g(x), g(t)) < \frac{\varepsilon}{2}.$$

Since E is dense in X, there is $t \in B_\delta(x) \cap E$. Now,

$$d(f(x), g(x)) \leq d(f(x), f(t)) + \underbrace{d(f(t), g(t))}_{=0} + d(g(t), g(x)) < \varepsilon = d(f(x), g(x))$$

It is Contradiction.

「Thm. Let $f: X \to Y$ be a map defined on a Compact Metric space X, and any metric space Y.

If f is Continuous, then it is Continuous Uniformly.

Proof. Suppose f is continuous. Given $\epsilon > 0$, for any $x \in X$, there exists $\delta_x > 0$ s.t

$$t \in B_{\delta_x}(x) \Rightarrow f(t) \in B_{\frac{\epsilon}{2}}(f(x))$$

Since the set $\{ B_{\frac{\delta_x}{2}}(x) \mid x \in X \}$ is open covering of X and X is Compact, there exists a finite subcover of X:

$$\{ B_{\frac{\delta_i}{2}}(x_i) \mid i = 1, 2, \cdots, n \} \quad \text{where } \delta_i = \delta_{x_i}.$$

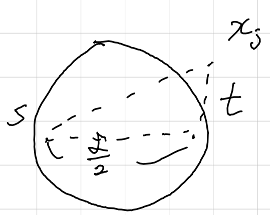
Take $\delta = \min\{\delta_1, \cdots, \delta_n\}$. Choose $s, t \in X$ s.t $d(s,t) < \frac{\delta}{2}$. (Then, $\delta_i + \delta \leq 2\delta_i$, $\forall i$)

Since the existence of finite subcover, for some $1 \leq j \leq n$

$$d(t, x_j) < \frac{\delta_j}{2}, \text{ which implies } d(s, x_j) \leq d(s,t) + d(t, x_j) < \frac{\delta}{2} + \frac{\delta_j}{2} \leq \delta_j$$

Finally,

$$d(f(s), f(t)) \leq d(f(s), f(x_j)) + d(f(x_j), f(t)) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$



「Section 4. 7. Define

$$f : \mathbb{R}^2 \to \mathbb{R} : (x,y) \mapsto \begin{cases} 0 & \text{if } (x,y)=(0,0) \\ \dfrac{xy^2}{x^2+y^4} & \text{otherwise} \end{cases}, \quad g : \mathbb{R}^2 \to \mathbb{R} : (x,y) \mapsto \begin{cases} 0 & \text{if } (x,y)=(0,0) \\ \dfrac{xy^2}{x^2+y^6} & \text{otherwise} \end{cases}$$

Show that: f is bounded on $\mathbb{R}^2$, g is not bounded in every neighborhood of $(0,0)$,
f is not continuous at $(0,0)$,
The restrictions of f, g on any straight line in $\mathbb{R}^2$ are continuous.

proof.
g is unbounded because: for each $x \neq 0$, take $y = x^{\frac{1}{3}}$. Then,

$$g(x,y) = \frac{x \cdot x^{\frac{2}{3}}}{x^2 + x^{6 \cdot \frac{1}{3}}} = \frac{x^{\frac{5}{3}}}{2x^2} = \frac{1}{2} \cdot \frac{1}{x^{\frac{1}{3}}} \to \infty \quad \text{as} \quad x \to 0.$$

Therefore unbounded.

$$2\sqrt{xy} \leq x+y \qquad (\sqrt{x}-\sqrt{y})^2 = x+y-2\sqrt{xy}$$

f is bounded, because: $2\sqrt{x^2y^4} = 2|x|\cdot|y|^2 \leq x^2+y^4$, that is

$$\left| \frac{xy^2}{x^2+y^4} \right| \leq \frac{1}{2}.$$ (But, how can I find the bound in general setting?)

To show the discontinuity of f at $(0,0)$, take for each $x \neq 0$, $y = \sqrt{x}$. Then.

$$f(x,y) = \frac{xy^2}{x^2+y^4} = \frac{x^2}{x^2+x^2} = \frac{1}{2} \neq f(0,0).$$

Therefore, $\left(\frac{1}{n}, \frac{1}{\sqrt{n}}\right) \to (0,0)$, But $f\left(\frac{1}{n}, \frac{1}{\sqrt{n}}\right) \to \frac{1}{2}$, contradiction.

To show the last assertion, if $x = c$ for some $c \in \mathbb{R}$,
$y \mapsto \dfrac{cy^2}{c^2+y^4}$ is clearly continuous, similarly $y \mapsto \dfrac{cy^2}{c^2+y^6}$.

If $y = ax$ for some $a \in \mathbb{R}$, then

$$x \mapsto \frac{a^2x^3}{x^2+a^4x^8} \quad \text{and} \quad x \mapsto \frac{a^2x^3}{x^2+a^6x^{12}} \quad \text{are continuous.}$$